STAT 578: Spectral Methods, Network Analysis, and Nonconvex Optimization

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1 Model and Setup

We consider the Stochastic Block Model (SBM):

$$A \sim \mathrm{SBM}(C, B), \qquad B \in [0,1]^{k\times k},$$

where $C \in \{0,1\}^{n\times k}$ is the community membership matrix and B is the block probability matrix.

We parameterize

$$B = \rho_n B_0, \qquad \max_i (B_0)_{ii} = 1,$$

with ρ_n a sparsity parameter. Define

$$\lambda_{\min} := \lambda_k(B_0),$$

the smallest nonzero eigenvalue of B_0. This governs the closeness of clusters.

Example 1.1 (Two-block symmetric case).

$$B = \begin{bmatrix} a & b \\ b & a \end{bmatrix}, \qquad \rho_n = a, \qquad B_0 = \begin{bmatrix} 1 & b/a \\ b/a & 1 \end{bmatrix}, \qquad \lambda_{\min} = 1 - \tfrac{b}{a}.$$

2 Information-Theoretic and Algorithmic Bounds

2.1 Lower and Upper Bounds

Lower bound. For any estimator $\hat{C}$,

$$\inf_{\hat{C}} \sup \ \mathbb{E}\big[\ell(\hat{C}, C)\big] \ \gtrsim \ \exp\!\big(-c\lambda_{\min}^2 n\rho_n\big)\,.$$

Upper bound. By Davis–Kahan perturbation and refinement arguments ([Lei and Rinaldo, 2015]),

$$\ell(\hat{C}, C) \ \lesssim \ \frac{c_k}{\lambda_{\min}\sqrt{n\rho_n}} \quad \text{w.h.p.}$$

Thus the threshold for exact recovery is

$$n\rho_n\lambda_{\min}^2 \gg \log n \qquad \Rightarrow \qquad \mathbb{P}[\hat{C} = C] \to 1.$$

2.2 Interpretation

- Lower bound: $\exp(-\lambda_{\min}^2 n\rho_n)$.
- Upper bound: $1/(\lambda_{\min}\sqrt{n\rho_n})$.

For fixed k, the information-theoretic and algorithmic rates align.

2.3 Algorithmic Realization

[Zhang and Zhou, 2015] provide a polynomial-time algorithm that achieves the optimal recovery threshold. The pipeline is

$$A \longrightarrow \hat{U} \in \mathbb{R}^{n\times k} \xrightarrow{\text{cluster rows via } k\text{-means}} \hat{C} \in \{0,1\}^{n\times k}.$$

With an additional likelihood refinement step,

$$\hat{\hat{C}} \in \{0,1\}^{n\times k}, \qquad \mathbb{P}[\hat{\hat{C}} = C] \to 1.$$

3 Exact Recovery of the Refined Estimator

We now show that the refined estimator $\hat{C}$ achieves exact recovery with high probability, not just approximate recovery.

For any clustering estimator $\hat{C}$,

$$\Pr\big[\ell(\hat{C}, C) \geq \tfrac{1}{n}\big] \;\leq\; n\,\mathbb{E}[\ell(\hat{C}, C)].$$

If Davis–Kahan plus spectral perturbation yield

$$\mathbb{E}[\ell(\hat{C}, C)] \;\lesssim\; \exp(-C\lambda_{\min}^2 n\rho_n),$$

then

$$\Pr\big[\ell(\hat{C}, C) \geq \tfrac{1}{n}\big] \;\leq\; n\exp(-C\lambda_{\min}^2 n\rho_n).$$

If

$$\lambda_{\min}^2 n\rho_n \;\geq\; C_0 \log n,$$

then

$$n\exp(-C\lambda_{\min}^2 n\rho_n) \;\leq\; n^{-100},$$

for sufficiently large C_0. Thus, with high probability,

$$\Pr[\tilde{C} = C] \;\geq\; 1 - n^{-100}.$$

Example 3.1 (Two-block case: population eigenvectors). For

$$B = \begin{bmatrix} a & b \\ b & a \end{bmatrix}, \qquad \text{with } \tfrac{n}{2} \text{ nodes per community},$$

the eigenvectors are

$$u_1 = \tfrac{1}{\sqrt{n}} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}, \qquad u_2 = \tfrac{1}{\sqrt{n}} \begin{bmatrix} 1 \\ \vdots \\ 1 \\ -1 \\ \vdots \\ -1 \end{bmatrix}.$$

Remark 3.2. Compute the leading eigenvectors $\hat{U}$ and use the sign of the second eigenvector:

$$\hat{C}_i = \mathbf{1}\{\hat{u}_{2,i} \geq 0\}.$$

We want to show

$$\operatorname{sign}(\hat{u}_2) = \operatorname{sign}(u_2) \quad \text{w.h.p.}$$

This holds provided

$$\|\hat{u}_2 - u_2\|_\infty \;\le\; \min_i |(u_2)_i|.$$

Since $(u_2)_i = \pm 1/\sqrt{n}$, this reduces to

$$\|\hat{u}_2 - u_2\|_\infty \;\le\; \tfrac{1}{\sqrt{n}}.$$

In general, if Δ is the minimum separation between cluster centers in the spectral embedding, then exact recovery follows once

$$\inf_{O \in \mathbb{O}^k} \max_{i \in [n]} \|(\hat{U}O - U)_{i,:}\|_2 \;\ll\; \Delta.$$

4 Entrywise Perturbation and Incoherence

Definition 4.1 (Incoherence)**.** For an orthonormal matrix $U \in \mathbb{R}^{n \times r}$, the incoherence parameter μ is the smallest number such that

$$\max_{i \in [n]} \|U_{i,:}\|_2 \;\le\; \sqrt{\tfrac{\mu r}{n}}.$$

Equivalently,

$$\|U\|_{2,\infty} := \max_{i \in [n]} \|U_{i,:}\|_2 \;\le\; \sqrt{\tfrac{\mu r}{n}}.$$

Examples.

- For $r = 1$, if $u_1 = e_1 = (1, 0, \ldots, 0)^\top$, then $\|u_1\|_{2,\infty} = 1$, so $\mu = \sqrt{n}$.
- If $u_2 = (1/\sqrt{n}, \ldots, 1/\sqrt{n})$, then $\|u_2\|_{2,\infty} = 1/\sqrt{n}$, so $\mu = 1$.

Example matrices.

$$M_1 = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad M_2 = \tfrac{1}{n} \begin{bmatrix} \lambda & \lambda & \lambda \\ \lambda & \lambda & \lambda \\ \lambda & \lambda & \lambda \end{bmatrix}.$$

Remark 4.2 (SBM incoherence). For SBMs with minimum community size $n_{\min}$, one has $\mu \approx n/n_{\min}$.

Remark 4.3 (Asymmetric case). If $M \in \mathbb{R}^{d_1 \times d_2}$, embed into $\begin{bmatrix} 0 & M \\ M^\top & 0 \end{bmatrix}$ and use $U_{\text{new}} = \frac{1}{\sqrt{2}} \begin{bmatrix} U \\ V \end{bmatrix}$ when $M = U\Sigma V^\top$. This reduction is valid provided d_1/d_2 is not too unbalanced.

Theorem 4.4 (Entrywise perturbation bound)**.** *Suppose*

$$\hat{M} = M + N, \qquad M = U\Lambda U^\top, \quad M \in \mathbb{R}^{n \times n} \text{ symmetric.}$$

Assume:

- $U \in \mathbb{R}^{n \times r}$ *has incoherence parameter* μ*,*
- $\lambda = \min_i |\lambda_i|$*,*
- $N_{ij} = N_{ji}$ *are i.i.d. sub-Gaussian with variance proxy* σ^2*,*
- $\lambda/\sigma \;\ge\; C_0\sqrt{n \log n}$*.*

Then with probability at least $1 - n^{-10}$*,*

$$\inf_{O \in \mathbb{O}^r} \|\hat{U}O - U\|_{2,\infty} \;\le\; C\frac{\sigma\mu\sqrt{r \log n}}{\lambda}.$$

Comparison with Davis–Kahan.

- Davis–Kahan:

$$\|\hat{U} - UO\|_F \;\lesssim\; \frac{C\sigma\sqrt{rn}}{\lambda}.$$

- Entrywise refinement:

$$\max_i \|(\hat{U} - UO)_{i,:}\|_2 \;\le\; C\frac{\mu\sigma\sqrt{r\log n}}{\lambda}.$$

Thus entrywise perturbation bounds are strictly stronger and are needed for exact cluster recovery, since they control individual coordinates.

4.1 Proof sketch of the entrywise perturbation bound

Proof sketch. We want to bound a single row error:

$$(\hat{U} - UO)_{i,:} \;=\; e_i^\top(\hat{U} - UO),$$

where $O = \arg\min_{O\in\mathbb{O}^r} \|\hat{U} - UO\|_F$.

Let $M = U\Lambda U^\top$, $\hat{M} = M + N$. Then

$$U^\top U = I_r, \quad \hat{U}^\top\hat{U} = I_r.$$

By spectral decomposition,

$$\hat{U} = \hat{M}\hat{U}\hat{\Lambda}^{-1},$$

provided $\hat{\Lambda}$ is invertible. This holds since, by Weyl's inequality and matrix concentration,

$$\min_{k\le r} |\hat{\lambda}_k| \ge c\lambda, \qquad \max_i |\hat{\lambda}_i| \le C\sigma\sqrt{n}, \qquad \|N\| \le C\sigma\sqrt{n}$$

with high probability.

$$e_i^\top(\hat{U} - UO) = e_i^\top(I - UU^\top)\hat{M}\hat{U}\hat{\Lambda}^{-1} + e_i^\top U(U^\top\hat{U} - O).$$

The second term is small since O aligns subspaces optimally. Focusing on the first term:

$$e_i^\top(I - UU^\top)\hat{M}\hat{U}\hat{\Lambda}^{-1} = e_i^\top(I - UU^\top)(M + N)\hat{U}\hat{\Lambda}^{-1}.$$

Since $(I - UU^\top)M = 0$, this reduces to

$$e_i^\top(I - UU^\top)N\hat{U}\hat{\Lambda}^{-1}.$$

Thus

$$e_i^\top(\hat{U} - UO) = e_i^\top N\hat{U}\hat{\Lambda}^{-1} - e_i^\top UU^\top N\hat{U}\hat{\Lambda}^{-1} + (\text{smaller terms}).$$

Bounding:

$$\|e_i^\top(\hat{U} - UO)\| \;\le\; \frac{\|e_i^\top N\hat{U}\|}{\lambda/2}.$$

The key quantity is

$$\|e_i^\top N\hat{U}\|.$$

And the rest of the quantity will show below. If $r = 1$ and $\hat{u}$ were independent of N, then $\sum_j N_{ij}\hat{u}_j$ would be sub-Gaussian with variance proxy σ^2, yielding $\|e_i^\top N\hat{u}\| = O(\sqrt{\log n})$. In reality, $\hat{U}$ depends on N, so more delicate leave-one-out arguments are required.

Recall the decomposition

$$e_i^\top(\hat{U} - UO) = e_i^\top N\hat{U}\hat{\Lambda}^{-1} \;+\; T_2 \;+\; T_3,$$

where

$$T_2 = -e_i^\top UU^\top N\hat{U}\hat{\Lambda}^{-1}, \qquad T_3 = e_i^\top U(U^\top\hat{U} - O).$$

Bounding T_2. Using $\|AB\| \le \|A\|\|B\|$ and $\|U^\top e_i\|_2 \le \|U\|_{2,\infty}$,

$$\|T_2\| = \|e_i^\top UU^\top N\hat{U}\hat{\Lambda}^{-1}\| \;\le\; \|e_i^\top U\|_2\,\|U^\top N\hat{U}\|\,\|\hat{\Lambda}^{-1}\| \;\le\; \|U\|_{2,\infty}\,\|N\|\,\tfrac{2}{\lambda}.$$

Since $\|U\|_{2,\infty} \le \sqrt{\mu r/n}$ and $\|N\| \lesssim \sigma\sqrt{n}$ w.h.p.,

$$\|T_2\| \;\lesssim\; \frac{\sqrt{\mu r}}{\sqrt{n}} \cdot \frac{\sigma\sqrt{n}}{\lambda} = \frac{\sigma\sqrt{\mu r}}{\lambda}.$$

Thus T_2 is lower-order compared to the leading term.

Bounding T_3. We need to control

$$T_3 = e_i^\top U(U^\top\hat{U} - O).$$

Note that O is chosen to minimize $\|\hat{U} - UO\|_F$. This optimization is equivalent to

$$O = \arg\max_{O\in\mathbb{O}^r} \mathrm{Tr}(\hat{U}^\top UO).$$

Writing the SVD of $U^\top\hat{U}$ as $W\Sigma V^\top$, we have $O = VW^\top = I$, and

$$\|U^\top\hat{U} - O\| = \|\Sigma - I\| = \max_j |1 - \cos\theta_j| \;\le \|\sin\Theta(U,\hat{U})\|^2,$$

where θ_j are principal angles. By Davis–Kahan,

$$\|\sin\Theta(U,\hat{U})\| \;\lesssim\; \frac{\|N\|}{\lambda}.$$

Hence

$$\|U^\top\hat{U} - O\| \;\lesssim\; \Big(\frac{\|N\|}{\lambda}\Big)^2.$$

Therefore,

$$\|T_3\| \;\le\; \|e_i^\top U\|_2\,\|U^\top\hat{U} - O\| \;\lesssim\; \|U\|_{2,\infty} \cdot \frac{\|N\|^2}{\lambda^2}.$$

□

We now bound the key term without resorting to the naive Davis–Kahan estimate.

$$\|e_i^\top N\hat{U}\hat{\Lambda}^{-1}\| \;\le\; \frac{2}{\lambda}\,\|e_i^\top N\hat{U}\|.$$

Decompose

$$\|e_i^\top N\hat{U}\| \;\le\; \|e_i^\top NU\| \;+\; \|e_i^\top N(\hat{U} - UO)\|.$$

First term. Each row $e_i^\top N$ has i.i.d. sub-Gaussian entries with variance proxy σ^2. Since U is orthonormal, $e_i^\top NU$ is sub-Gaussian with the same proxy. By a union bound over r coordinates,

$$\|e_i^\top NU\| \lesssim \sigma\sqrt{r\log n}, \qquad \text{w.h.p.}$$

Second term. We have

$$\|e_i^\top N(\hat{U} - UO)\| \le \|e_i^\top N\|\,\|\hat{U} - UO\|.$$

Concentration gives $\|e_i^\top N\| \lesssim \sigma\sqrt{n}$, and Davis–Kahan gives $\|\hat{U} - UO\| \lesssim \|N\|/\lambda \lesssim \sigma\sqrt{n}/\lambda$. Therefore

$$\|e_i^\top N(\hat{U} - UO)\| \lesssim \frac{\sigma^2 n}{\lambda}.$$

Combined bound. Altogether,

$$\|e_i^\top N\hat{U}\| \lesssim \sigma\sqrt{r\log n} + \frac{\sigma^2 n}{\lambda},$$

and hence

$$\|e_i^\top N\hat{U}\hat{\Lambda}^{-1}\| \lesssim \frac{\sigma\sqrt{r\log n}}{\lambda} + \frac{\sigma^2 n}{\lambda^2}.$$

4.2 Leave-one-out construction

Fix $i \in [n]$ and let $n_i^\top := e_i^\top N$ be the i-th noise row. Decompose

$$\|e_i^\top N\hat{U}\| \le \|e_i^\top N(\hat{U}O - \hat{U}^{(i)})\| + \|e_i^\top N\hat{U}^{(i)}\|,$$

where $\hat{U}^{(i)}$ are the top-r eigenvectors of the leave-one-out matrix

$$M^{(i)} := M + (I - e_i e_i^\top)N(I - e_i e_i^\top).$$

Explicitly,

$$M^{(i)} = M + N^{(i)}, \qquad N^{(i)} := N - e_i n_i^\top - n_i e_i^\top + N_{ii} e_i e_i^\top.$$

Thus $M^{(i)}$ has the same distribution as $M + N$ but with the i-th row/column of N zeroed out, hence n_i is independent of $\hat{U}^{(i)}$.

Bounding the independent part. Since n_i is independent of $\hat{U}^{(i)}$, by a matrix Hoeffding/Bernstein inequality,

$$\|e_i^\top N\hat{U}^{(i)}\| \lesssim \sigma\sqrt{r\log n} \quad \text{w.h.p.}$$

Bounding the remainder. We next control $\|\hat{U}O - \hat{U}^{(i)}\|$. By Weyl's inequality, the eigenvalues of $M^{(i)}$ are still well-separated:

$$\min_{j\le r} |\hat{\lambda}_j^{(i)}| \ge \tfrac{1}{4}\lambda, \qquad \max_j |\hat{\lambda}_j^{(i)}| \le C\sigma\sqrt{n}.$$

Now by Davis–Kahan,

$$\|\sin\Theta(\hat{U}, \hat{U}^{(i)})\| \lesssim \frac{\|(M+N) - (M+N^{(i)})\|}{\lambda} = \frac{\|e_i n_i^\top + n_i e_i^\top - N_{ii} e_i e_i^\top\|}{\lambda}.$$

This perturbation has operator norm $\lesssim \|n_i\|_2$, so

$$\|\sin\Theta(\hat{U}, \hat{U}^{(i)})\| \lesssim \frac{\|n_i\|_2}{\lambda}.$$

Remainder contribution. Hence

$$\|e_i^\top N(\hat{U}O - \hat{U}^{(i)})\| \;\le\; \|n_i\|_2 \,\|\hat{U}O - \hat{U}^{(i)}\| \;\lesssim\; c\sigma\sqrt{n}\|\sin\Theta(\hat{U}, \hat{U}^{(i)})\|.$$

Since $\|n_i\|_2^2 \lesssim \sigma^2 n$ w.h.p.

4.3 Summary of the leave-one-out bound

Putting the two parts together,

$$\|e_i^\top N\hat{U}\| \;\lesssim\; \sigma\sqrt{r\log n} \;+\; \frac{\sigma^2 n}{\lambda} \qquad \text{(w.h.p. for this fixed } i\text{)}.$$

$$\max_{i\in[n]} \|e_i^\top N\hat{U}\| \;\lesssim\; \sigma\sqrt{r\log n} \;+\; \frac{\sigma^2}{\lambda}\Big(n\sqrt{\log n}\|\hat{U}\|_{2,\infty}\Big) \qquad \text{(w.h.p.)}$$

This refines the earlier bound and shows how the leave-one-out trick controls dependence between N and $\hat{U}$.

4.4 Entrywise perturbation: final per-row bound

We begin with

$$\|e_i^\top N\hat{U}\hat{\Lambda}^{-1}\| \;\le\; \frac{2}{\lambda}\,\|e_i^\top N\hat{U}\|.$$

Now, to control $\|\hat{U}\|_{2,\infty}$, we compare $\hat{U}$ with its leave-one-out analogue $\hat{U}^{(i)}$. By the triangle inequality,

$$\|\hat{U}^{(i)}\|_{2,\infty} \;\le\; \|\hat{U}^{(i)}O - \hat{U}\|_{2,\infty} + \|\hat{U}\|_{2,\infty}.$$

By the entrywise Davis–Kahan bound,

$$\|\hat{U}^{(i)}O - \hat{U}\|_{2,\infty} \;\le\; \|\sin\Theta(\hat{U}^{(i)}, \hat{U})\|_F + \|\hat{U}\|_{2,\infty} \;\lesssim\; \frac{C\sigma\sqrt{n\log n}}{\lambda}\,\|\hat{U}^{(i)}\|_{2,\infty} + \|\hat{U}^{(i)}\|_{2,\infty}.$$

Hence,

$$\|\hat{U}\|_{2,\infty} \;\le\; \frac{C\sigma\sqrt{n\log n}}{\lambda}\,\|\hat{U}^{(i)}\|_{2,\infty} + 2\|\hat{U}^{(i)}\|_{2,\infty}.$$

By a standard leave-one-out comparison,

$$\|\hat{U}^{(i)}\|_{2,\infty} \;\le\; \tfrac{1}{2}\|\hat{U}^{(i)}\|_{2,\infty} + \|\hat{U}\|_{2,\infty},$$

which yields

$$\|\hat{U}^{(i)}\|_{2,\infty} \;\le\; \frac{C\sigma\sqrt{n\log n}}{\lambda}\,\|\hat{U}\|_{2,\infty}.$$

Finally, plugging this back into the bound for $\|e_i^\top(\hat{U} - UO)\|$ gives

$$\|e_i^\top(\hat{U} - UO)\| \;\le\; \frac{C\sigma\sqrt{n}}{\lambda}\,\|U\|_{2,\infty} + \frac{C\sigma\sqrt{n\log n}}{\lambda}\,\|\hat{U}\|_{2,\infty} \;\lesssim\; \frac{C\sigma\sqrt{n\log n}}{\lambda}\,\|U\|_{2,\infty}.$$

5 Application of exact clustering

1. **SBM model:** For $i < j$, the edges are independent with

$$A_{ij} \sim \mathrm{Bern}\big(B_{c_i c_j}\big)\,, \qquad C : [n] \to [k].$$

2. **Parameterization:** $B = \rho_n B_0$ with $\max_i (B_0)_{ii} = 1$.

3. **Separation:** $\lambda_{\min}(B_0) \geq \lambda_0 > 0$ (absolute constant).

4. **Sparsity:** $n\rho_n \geq C_0 \log n$ for a sufficiently large constant C_0.

Let $\hat{U} \in \mathbb{R}^{n \times k}$ be the matrix of top-k eigenvectors of A. Then, with probability at least $1 - n^{-10}$, applying k-means to the rows of $\hat{U}$ returns the true communities exactly up to a permutation of labels.

Remark 5.1. For Erdős–Rényi graphs, the condition $n\rho_n$ relative to $\log n$ governs connectivity: if $n\rho_n \ll \log n$, the graph is disconnected almost surely, while if $n\rho_n \to \infty$, then with high probability the graph contains a giant connected component.

References

J. Lei and A. Rinaldo. Consistency of spectral clustering in stochastic block models. *The Annals of Statistics*, 43(1), Feb. 2015. ISSN 0090-5364. doi: 10.1214/14-aos1274. URL http://dx.doi.org/10.1214/14-AOS1274.

A. Y. Zhang and H. H. Zhou. Minimax rates of community detection in stochastic block models, 2015. URL https://arxiv.org/abs/1507.05313.